

SAHITHI RUPIREDDY

Software Engineer

Email: rupireddysahithi14@gmail.com | Phone: (314) 861-8469 | Location: Saint Louis, MO | [LinkedIn](#)

SUMMARY

- Software Engineer with around 4 years of experience in designing, developing, and optimizing scalable applications across various industries.
- Expertise in backend and frontend development, specializing in Java, Python, TypeScript, and JavaScript with frameworks like Spring Boot, Node.js, React.js, and Angular.
- Skilled in microservices architecture, event-driven systems, and cloud computing, leveraging AWS, Azure, and GCP for scalable and efficient deployments.
- Experienced in implementing LLM-powered applications, AI-driven automation, and machine learning models using OpenAI API, TensorFlow, and Scikit-learn.
- Strong knowledge of authentication and security protocols including OAuth2, JWT, OpenID Connect, and SSL/TLS for robust data protection.
- Proficient in database management and optimization, working with MySQL, PostgreSQL, MongoDB, and Redis to enhance query performance and data handling.
- Expertise in CI/CD, DevOps, and containerization, utilizing Docker, Kubernetes, Terraform, GitHub Actions, and Jenkins for seamless automation and deployment.
- Adept at API development and integration, working with RESTful APIs, GraphQL, and AWS API Gateway to ensure interoperability across systems.
- Strong advocate of agile development, utilizing Scrum, Kanban, and TDD/BDD methodologies to drive efficient software delivery.
- Passionate about problem-solving, optimizing software performance, and leveraging the latest technologies to build high-impact solutions.

SKILLS

- **Programming Languages:** Java, Python, TypeScript, SQL, JavaScript
- **Frontend Development:** HTML5, CSS3, React.js, Angular, Vue.js, Bootstrap, Material-UI, Tailwind CSS
- **Backend Development:** Spring Boot, Spring Cloud, Node.js, Express.js, Django, Flask
- **Database Management:** MySQL, PostgreSQL, MongoDB, SQL Server, Redis, Hibernate, JPA
- **Messaging & Event-Driven Architecture:** Apache Kafka, RabbitMQ
- **Version Control & CI/CD:** Git, GitHub, GitLab CI, Jenkins, GitHub Actions, Bitbucket
- **Containerization & Orchestration:** Docker, Kubernetes, Docker Compose, Helm
- **Authentication & Security:** JWT, OAuth2, OpenID Connect, SSL/TLS, RBAC
- **Cloud & DevOps:** AWS (EC2, Lambda, S3, RDS, DynamoDB), Azure, GCP, Terraform, Ansible
- **API Development & Management:** RESTful APIs, GraphQL, SOAP APIs, AWS API Gateway, Postman, Swagger
- **Testing & Automation:** JUnit, Mocha, Jest, Cypress, Selenium, Postman
- **Machine Learning & AI:** TensorFlow, Scikit-learn, OpenAI API, LLM integration (GPT, BERT)
- **Agile & Software Development:** Scrum, Kanban, TDD, BDD, Jira, Confluence
- **Miscellaneous Tools:** SonarQube, IntelliJ IDEA, Visual Studio Code, Draw.io, Slack

WORK EXPERIENCE

Molina Healthcare

Feb 2025 – Present

Software Engineer

- Designed and implemented a scalable API Gateway for Healthcare Interoperability using GraphQL, AWS API Gateway, and Spring Cloud, ensuring compliance with FHIR and HL7 interoperability standards for seamless healthcare data exchange.
- Developed secure authentication and authorization mechanisms within API Gateway for Healthcare Interoperability using OAuth2, OpenID Connect, and JWT, enhancing data security and access control for integrated healthcare applications.
- Optimized API response times in API Gateway for Healthcare Interoperability by integrating Redis caching and event-driven architecture with Apache Kafka, reducing latency by 20% and improving real-time patient data access.

Molina Healthcare (Internship)

July 2024 – Dec 2024

Software Engineer

- Developed and deployed Automated Insurance Eligibility Verification using Spring Boot, Flask, and SQL Server, reducing manual verification time and ensuring real-time eligibility checks for healthcare providers.
- Designed and implemented RESTful APIs with AWS API Gateway to securely retrieve patient eligibility data from multiple insurance providers, improving interoperability.
- Integrated JWT-based authentication and SSL/TLS encryption to enhance security and compliance with HIPAA regulations in insurance data processing.
- Optimized database queries in MySQL and SQL Server, reducing query execution time by **25%**, improving the efficiency of insurance verification processes, and enhancing system performance.
- Automated the deployment pipeline using GitHub Actions, Jenkins, and Docker, ensuring seamless CI/CD integration for microservices handling insurance eligibility requests.

- Conducted rigorous testing with JUnit, Mocha, and Postman, ensuring high availability and accuracy of the insurance verification system before production rollout.

Trigent Software
Software Engineer

Feb 2020 – Dec 2022

- Led the Legacy Monolith to Microservices Migration, refactoring a monolithic enterprise application into a Spring Boot and Node.js microservices architecture, improving system scalability and maintainability.
- Designed and implemented 30+ RESTful APIs using Express.js and Spring Boot, replacing outdated SOAP APIs to enhance system integration and data exchange.
- Optimized database performance by migrating legacy MySQL tables to MongoDB and Hibernate, reducing query execution time and improving data retrieval efficiency.
- Developed event-driven communication using RabbitMQ, ensuring seamless inter-service messaging and enhancing the reliability of distributed systems.
- Containerized microservices with Docker and Helm, enabling efficient deployment, scaling, and orchestration across cloud-based environments.
- Integrated OAuth2 and SSL/TLS for secure authentication and authorization, strengthening access control and compliance with industry security standards.
- Automated CI/CD pipelines using GitLab CI and Jenkins, reducing deployment time and ensuring smooth application rollouts with minimal downtime.
- Conducted rigorous testing using JUnit and Selenium, improving code quality and ensuring application stability before transitioning services to production.

PROJECTS

LLM-Powered Chatbot for Personalized Learning

- Built a custom AI tutor chatbot leveraging GPT-4 fine-tuned with LangChain and Pinecone vector databases, providing adaptive learning experiences.
- Integrated speech-to-text and text-to-speech capabilities using Whisper AI and ElevenLabs, enabling voice-based interactions for a seamless user experience.
- Developed an interactive web interface using Next.js and Tailwind CSS, offering a dynamic and user-friendly learning environment.

E-Commerce Price Optimization Engine

- Built a dynamic pricing model using XGBoost and time series forecasting to adjust product prices based on demand, competitor trends, and inventory levels.
- Designed an API-driven microservices backend with FastAPI, PostgreSQL, and Redis, enabling real-time price updates across multiple e-commerce platforms.
- Deployed the model on AWS Lambda and S3, ensuring seamless integration with cloud-based e-commerce platforms for automated pricing adjustments.

EDUCATION

Master of Science (M.S.) in Computer Science

Jan 2023 - Dec 2024

Saint Louis University, Saint Louis, Missouri, USA | GPA: 3.64/4.00

Bachelor of Technology (B.Tech.) in Computer Science

Jun 2017 - Aug 2021

Amrita School of Engineering, Coimbatore, India | GPA: 7.45/10.00